



Álvaro Yángüez Bachiller

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Address: 4 place Jussieu, 75252 , Paris, France (Work)

EDUCATION AND TRAINING

01/10/2023 – 30/09/2026 Paris, France

PHD STUDENT Sorbonne Université

Doctoral candidate inside the Marie Curie Doctoral Network "Quantum Safe Internet".

12/10/2021 – 10/09/2023 Munich, Germany

M.SC. QUANTUM SCIENCE AND TECHNOLOGY TUM and LMU

I took the following courses:

- Quantum Information* (Prof. Dr. Igancio Cirac)
- Quantum Optics* (Prof. Dr. Immanuel Bloch)
- Quantum Statistical Inference* (Prof. Dr. Michael Wolf)
- Tensor Networks* (Prof. Dr. Jan von Delft)
- Quantum Communication* (Prof. Dr. Harald Weinfurter)
- Quantum Hardware* (Prof. Dr. Monika Aidelsburger)

Thesis Quantum tomography enabled by homogeneous Markovian evolutions (supervised by Prof. Dr. Michael Wolf)

17/09/2017 – 08/02/2022 Madrid, Spain

DEGREE IN PHYSICS Universidad Complutense de Madrid

09/09/2015 – 20/05/2017 Madrid, Spain

INTERNATIONAL BACCALAUREATE IES Ramiro de Maeztu

09/09/2015 – 23/06/2017 Madrid, Spain

SPANISH BACCALAUREATE (IN SCIENCE) IES Ramiro de Maeztu

WORK EXPERIENCE

RESEARCH COLLABORATOR – HOLGER BOCHER-TUM – 01/04/2022 – 01/09/2023 – MUNICH, GERMANY

Theoretical research in the area of Quantum Communication, specifically in the problem of Remote State Preparation.

PUBLICATIONS

[4] Á. Yángüez, N. Avidan, J. Kochanowski, T.A. Hahn, "Accessible Quantum Correlations under Complexity Constraints", arXiv preprint, arXiv:2604.15540, 2026

[3] Á. Yángüez, T.A. Hahn, J. Kochanowski, "Efficient Quantum Measurements: Computational Max- and Measured Rényi Divergences and Applications", IEEE Transactions on Information Theory 72(6), 4085–4114, 2026.

Accepted as a talk in the workshop **Quantum Resources 2026**

[2] A. B. Grilo, Á. Yángüez "Quantum pseudoresources imply cryptography" IACR Communications in Cryptology 2 (4), 2026

[1] E. Diamanti, A. B. Grilo, A. Innocenzi, P. Lefebvre, V. Yacoub, Á. Yángüez "A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions" Quantum 10, 2124, 2026.

● CONFERENCES

16/03/2026 – 21/03/2026

International Conference on Quantum Cryptography and Security (ICQCS)

I contributed with the talk *"A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions"*.

24/01/2026 – 30/01/2026

Quantum Information Processing (QIP) 2026

I presented the poster *"Efficient Quantum Measurements: Computational Max- and Measured Rényi Divergences and Applications"*.

12/05/2025 – 14/05/2025

QSI Workshop 2025

As part of the MSCA Doctoral Network "Quantum Safe Internet", I gave the talk *"Quantum Pseudoresources Imply Cryptography"*.

06/05/2025 – 08/05/2025

Workshop: Entanglement, Information and Complexity in Quantum Systems

I was invited to give the talk *"Quantum Pseudoresources Imply Cryptography"* at the Higgs Centre for Theoretical Physics.

02/09/2024 – 06/09/2024

14th international conference on quantum cryptography (QCRYPT2024)

Contributed with two posters:

- E. Diamanti, A. B. Grilo, A. Innocenzi, P. Lefebvre, V. Yacoub, Á. Yáñez *"A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions"*.
- A. Innocenzi, V. Yacoub, Á. Yáñez, P. Lefebvre, A. B. Grilo, E. Diamanti, *"Experimental implementation of Simulation secure Quantum Oblivious Transfer"*.

04/11/2024 – 08/11/2024

7th International Conference for Young Quantum Information Scientists (YQIS24)

Contributed with the poster:

- E. Diamanti, A. B. Grilo, A. Innocenzi, P. Lefebvre, V. Yacoub, Á. Yáñez *"A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions"*.

● INVITED SEMINARS

15/12/2025

"Efficient Quantum Measurements: Computational Max- and Measured Rényi Divergences and Applications" MathQI UCM

03/12/2025

"Efficient Quantum Measurements: Computational Max- and Measured Rényi Divergences and Applications" QAT Team-Inria

07/11/2025

"Efficient Quantum Measurements: Computational Max- and Measured Rényi Divergences and Applications" University of Cambridge

30/10/2025

"Efficient Quantum Measurements: Computational Max- and Measured Rényi Divergences and Applications" PhyQus Team-Inria

22/10/2025

"Efficient Quantum Measurements: Computational Max- and Measured Rényi Divergences and Applications" IRIF, Paris

05/01/2025

"A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions" TUM, Munich

29/01/2025

"On the role of pseudo-entanglement in cryptography" Jiq epiq 2025, Paris

18/06/2024

"A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions" QURIOSITY, Telecom Paris

17/05/2024

"A Practical Protocol for Quantum Oblivious Transfer from One-Way Functions" QuSoft Seminar, Amsterdam

● **SCHOOLS**

11/03/2024 – 15/03/2024

Spring School on Post-Quantum Cryptography, O Porto.

Organised by Kathrin Hövelmanns, Andreas Hülsing and Alexander May.

Link <https://pqc-spring-school.nl/>

29/01/2024 – 02/02/2024

Winter School on Quantum Cryptography, Padova.

Selected Speakers: Giuseppe Vallone, Paolo Villoresi, Mohsen Razavi, and Peter Brown.

Link <https://quantum-safeinternet.com/conferencias/school-on-quantum-cryptograghy-sqc-padova/>

01/09/2020 – 10/09/2020

Summer School on Quantum Computing: Software for Near Term Quantum Devices

Organised by Diego Porras (CSIC) in the UIMP in 2020.

06/08/2016 – 17/08/2016

Scientific campus about energy organised by FECYT and the Spanish Ministry of Science and Innovation : « Energía: pasado, presente y futuro ».

Universidad Rovira i Virgili

● **VISITS**

03/11/2025 – 07/11/2025

Ángela Capel - University of Cambridge

11/04/2025 – 18/04/2025

Renato Renner - ETH

31/03/2025 – 10/04/2025

Thomas Vidick and Rotem Arnon - EPFL

06/05/2024 – 27/05/2024

Christian Schaffner and Stacey Jefferey - QuSoft, CWI, Amsterdam

● **SERVICES**

Organizing committee

Intenational Conference on Quantum Cryptography and Security (ICQCS)
(More information available at <https://icqcs.sciencesconf.org/>)

Program committee

External reviewer

Peer-reviewed journals:

- Physical Review Letter
- Physical Review A
- npj Quantum Information

Conferences:

- Theoretical Quantum Computing (TQC) 2026
- Quantum Information Procession (QIP) 2025
- 57th Annual ACM Symposium on Theory of Computing (STOCS) 2025

OTHER ACTIVITIES

13/01/2026 – 01/06/2026

Supervision Bachelor Project

I am supervising the last year's bachelor research project of Léna Husse, Johanna Theophilos and Matéo Boury on Quantum Oblivious Transfer in the Quantum Random Oracle model.

01/06/2024 – 10/06/2025

Organizer QI LIP6 Seminars

During one year, I was in charge of organizing the weekly seminars of QI LIP6 together with Matilde Baroni.

Soft Skills Training

During the PhD, I have participated in the following soft skills trainings:

- **Science Communication:** Facilitated by Richard A. Fuchs, a science journalist and a media trainer, the event was orchestrated by the National Institute for Science Communication (NaWik). This workshop aimed to equip scientists with essential skills to effectively convey their research to the broader public, thereby bridging the gap between the scientific community and society at large.
- **Science Commercialization:** The lectures were offered by Mads Vad Kristensen from DTU Skylab, an Innovation Partner in the innovation hub of the Technical University of Denmark. He leverages his expertise to support students and researchers in transforming their innovative ideas into viable startups, fostering the commercialization of cutting-edge technologies.

01/09/2024 – 10/09/2025

Responsible of the cultural activities of resident's committee in Colegio de España, Paris.

I was the responsible of organizing the cultural activities in Colegio de España in París during the academic year 2024-2025. Among others, I organized the talks of Prof. Dr. Ignacio Cirac, Prof. Dr. Ángela Capel Cuevas and Luis Cercós, chief architect of the BPI at the Centre Pompidou.

10/09/2018 – 10/09/2022

Theatre director

Besides my academical experience, I have been the director of the El Barracón theatre company for four years.

HONOURS AND AWARDS

10/09/2021

IBM Quantum Excellence certificate on 2021 Qiskit Global Summer School on Quantum Machine Learning – IBM

18/05/2017

Second prize in the National Prize of Initiation on Technological Investigation – UAITE

14/05/2019

Best director award in the XXIII Certamen de Teatro Universitario – Universidad Complutense de Madrid

● **LANGUAGE SKILLS**

Mother tongue(s): **SPANISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
GERMAN	A2	A2	A2	A2	A2
FRENCH	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user